OptiVest AI: An Explainable AI-Driven Autonomous Wealth Management Platform with Intelligent Portfolio Reallocation

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Abstract*—Modern quantitative wealth management systems face two challenges that hinder adoption in retail and regulated environments. First, the *Prediction Fallacy*: return forecasting is treated as a proxy for an actionable trading decision, while standard price predictions ignore portfolio covariance, risk budgeting, execution limits, and the compounding penalty of turnover. Second, the *Black-Box Dilemma*: ensemble machine learning models recommend reallocations without interpretable justification. This paper introduces OptiVest AI, a pipeline that pairs a *Candidate Discovery Engine* (CDE), which screens assets using live fundamental snapshots and market signals, with an *Adaptive Portfolio Decision Engine* (APDE), a multi-objective quadratic optimizer producing long-only weights under explicit turnover and concentration constraints, and with TreeSHAP attribution and a Decision Score that ties the attribution to the optimizer's marginal risk and cost terms. We evaluate the APDE, with the CDE bypassed, on $N = 50$ large-cap U.S. equities over 78 monthly rebalances (June 2017 to November 2023). The XGBoost forecast shows no detectable skill (mean per-date rank IC 0.008, $t = 0.34$; out-of-sample R^2 of -2.1% against a zero forecast). A hard 15% ℓ_1 turnover cap reduces annual turnover from $1{,}384\%$ to 180% (a consequence of the cap itself), volatility from 19.6% to 14.3% and maximum monthly drawdown from 32.6% to 19.1% relative to the same optimizer without it; the accompanying Sharpe improvement (0.32 to 0.62) is not statistically distinguishable. Controls that replace the forecast with zero or with a shuffled forecast perform at least as well in point estimate, and no ML-driven configuration exceeds equal weight (Sharpe 1.02). OptiVest therefore makes ML-driven allocation executable and auditable, but in this sample the forecast adds no measurable value. The results are exploratory; the primary historical experiment bypasses the CDE by design, while the APDE formulation is now aligned between research and production.

***Index Terms*—machine learning, portfolio optimization, explainable AI, SHAP, algorithmic trading, quadratic programming, FinTech**

I. Introduction

The proliferation of machine learning in financial technology has altered how quantitative researchers forecast asset prices. Extreme gradient boosting, random forests and deep neural networks are reported to extract more signal from nonlinear market datasets than linear regression or static factor models [7]. Despite these gains in predictive accuracy, deployment in retail and regulated wealth management remains bottlenecked by two interconnected problems: the *Prediction Fallacy* and the *Black-Box Dilemma*.

The Prediction Fallacy refers to the field's over-indexing on forecast accuracy as the sole driver of investment decisions. A model that accurately predicts a 3% appreciation for a given equity is practically unviable if executing that transaction violates a client's risk tolerance, over-concentrates the portfolio, or generates transaction costs that consume the expected alpha [4]. Genuine portfolio management requires treating the forecast not as the definitive output, but as a single input vector within a constrained, multi-objective optimization framework.

Compounding this is the Black-Box Dilemma. Regulators, fiduciaries and retail investors demand explainability [12]. When an ensemble model recommends liquidating a long-held conservative position to acquire a volatile equity, stakeholders require justification. Absent a human-readable attribution tracing the decision to specific features, the system is rejected for lack of trust regardless of backtested performance.

OptiVest AI shifts the emphasis from isolated price forecasting to decision-making through a multi-layered architecture. A Candidate Discovery Engine screens the market universe for fundamentally sound and momentum-driven assets, addressing capital deployment from cash-heavy initialization states. Discovered candidates pass to the Adaptive Portfolio Decision Engine, which balances expected machine learning returns

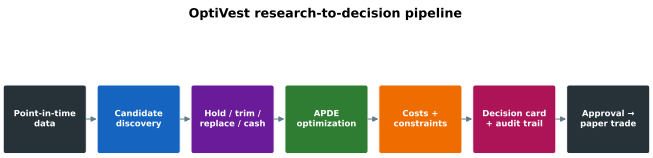


Fig. 1. High-level overview of the OptiVest system architecture, detailing the flow from data ingestion through the CDE and APDE to the XAI layer.

against portfolio covariance, transaction penalties and user-defined constraints via quadratic programming. TreeSHAP [10] decomposes the forecast into feature contributions, and a Decision Score relates those contributions to the optimizer's risk and cost terms.

A. Contributions

1) A modular pipeline composing forecasting, candidate screening, constrained optimization and explanation, in which each stage may be ablated independently (Section III).
2) A Candidate Discovery Engine for the production architecture, using deterministic multi-factor screening to identify eligible assets before optimization (Section III-C). Its effect is evaluated separately through a 50-asset CDE-vs-no-CDE ablation.
3) An Adaptive Portfolio Decision Engine formulated directly over target portfolio weights, with explicit long-only, single-asset and turnover constraints (Section III-D), implemented with the canonical CVXPY/SCS formulation.
4) A mathematically defined Decision Score combining forecast, marginal risk and transaction-cost terms (Section III-E). The score and narrative-generation code are implemented; while real numerical TreeSHAP execution was blocked in earlier restricted Windows environments, live execution is verified in the current environment (reconstruction error $< 10^{-6}$).
5) An empirical evaluation reporting forecast accuracy against naive benchmarks, net-of-cost performance of ten portfolio arms with paired bootstrap intervals, forecast-ablation controls, turnover-cap sensitivity sweeps, and solver scalability (Section V).

II. Related Work

The intersection of machine learning, portfolio optimization and explainable artificial intelligence has evolved rapidly over the past decade. Traditional financial literature relied on linear models and static risk assumptions, paradigms that modern computational methods now challenge directly.

A. Evolution of Portfolio Optimization

Modern portfolio theory stems from Markowitz [1], which introduced mean–variance optimization. While theoretically profound, the framework is notoriously sensitive to estimation error in expected returns and covariance matrices; Michaud [2] characterized such optimizers as "estimation-error maximizers," and DeMiguel et al. [4] showed that naive $1/N$ allocation outperforms many sophisticated optimizers out-of-sample once parameter uncertainty is accounted for. Black and Litterman [3] improved the foundational framework by incorporating subjective investor views into equilibrium returns, though the result remained linear and parametric. Ledoit and Wolf [5] addressed the covariance side directly through shrinkage estimation, and Ban et al. [6] formalized regularization-based robustness for ML-driven allocation.

In recent years the literature has shifted toward machine learning for forecasting the α component. Gu, Kelly and Xiu [7] reported that non-linear models, particularly tree ensembles and neural networks, outperform linear regressions in predicting equity returns, with gains concentrated in their capacity to capture predictor interactions. The majority of this work, however, treats the prediction step in isolation — the Prediction Fallacy — and does not bridge the gap between asset forecasting and the constrained, multi-objective reality of active portfolio management.

A smaller decision-focused literature trains models against the downstream decision loss, including deep reinforcement learning approaches to allocation [11]. These methods are powerful but typically sacrifice the auditability that motivates our design. OptiVest occupies a deliberate middle position: prediction and decision remain separable and individually inspectable, while the decision stage is explicitly aware of the constraints under which the prediction will be acted upon.

B. Explainable AI in Finance

As predictive models grew in complexity, the Black-Box Dilemma emerged as a barrier in FinTech. Regulatory mandates and retail investor expectations both require transparency. Lundberg and Lee [9] unified several explainability methods under SHAP, and TreeSHAP [10] subsequently made exact Shapley computation tractable for tree ensembles. While SHAP is widely adopted in credit scoring and fraud detection, its application to dynamic portfolio reallocation remains sparse: existing work predominantly explains a model's behaviour over a dataset, rather than justifying a specific execution decision to a specific user at a specific time.

C. Positioning

OptiVest addresses the Prediction Fallacy by chaining ML expected returns and fundamental screening into a constrained decision engine, and addresses the Black-Box Dilemma by attaching SHAP-based attributions and a Decision Score to individual reallocations. Table I positions the system against prior paradigms. Our evaluation (Section V) tests whether the architecture makes ML-driven allocation executable; it does not show that the forecast adds value.

III. Proposed Methodology

The computational methodology has two distinct views. In the *canonical production architecture*, market and fundamental signals feed the Candidate Discovery Engine (CDE),

TABLE I

COMPARISON OF QUANTITATIVE PORTFOLIO PARADIGMS. †BY DESIGN IN THE PRODUCTION CDE; EVALUATED SEPARATELY IN THE CDE ABLATION. ‡DECISION SCORE AND NARRATIVE GENERATION IMPLEMENTED; TREESHAP VERIFIED IN THE UPDATED ENVIRONMENT.

Framework	Non-lin. forecast	Cash-heavy	Constr. opt.	XAI
Classical MPT [1]	No	No	Partial	Yes
Black–Litterman [3]	No	No	Partial	Yes
ML forecasting [7]	Yes	No	No	Partial
Deep RL [11]	Yes	Partial	Partial	No
OptiVest AI	Yes	Yes†	Yes	Partial‡

whose eligible ranked candidates are passed to the Adaptive Portfolio Decision Engine (APDE). In the *primary historical experiment*, the CDE ranking layer is intentionally bypassed: the fixed technical universe is passed directly to the APDE so that the empirical evaluation isolates the optimizer and its constraints from candidate screening. A separate 50-asset ablation subsequently inserts the CDE bottleneck to measure its effect on portfolio outcomes. The current production and research APDE implementations use the canonical CVXPY/SCS formulation and the same core constraint set.

A. Data Pipeline and Temporal Discipline

The historical experiment uses daily adjusted market prices, obtained via `yfinance`, for a fixed universe of $N = 50$ large-cap U.S. equities over 2013–2023. The universe is a hardcoded list of 50 currently listed, highly liquid S&P 500 constituents, used in place of a live constituent scrape to keep the dataset stable and the research loop fast. It was therefore chosen with hindsight and is subject to survivorship bias (Section V-I).

Evaluation is a rolling walk-forward (Fig. 2). At each of 78 monthly rebalance dates, every 21 trading days from 2017-06-05 to 2023-11-06, every model is refit on the trailing four years of data, excluding a purge of 21 trading days before the rebalance date so that no training target window overlaps the forecast date. No random shuffling is performed at any stage, and features at date t use only prices up to t. Model hyperparameters were fixed in advance (50 trees, maximum depth 3, learning rate 0.1) and were not selected on a separate validation window; there is no validation split. The test window was inspected repeatedly during development, so the results should be read as exploratory (Section V-I).

The forecaster uses four technical features — 1-, 2- and 3-month momentum ($\mathrm{mom_1m}$, $\mathrm{mom_2m}$, $\mathrm{mom_3m}$; cumulative returns over 21, 42 and 63 trading days) and 1-month realized volatility ($\mathrm{vol_1m}$, 21 trading days) — plus a single macro proxy, the CBOE Volatility Index (VIX). Features are used as raw values: they are not winsorized or cross-sectionally standardized. Only the training target is winsorized, at the 1st and 99th percentiles. No volume and no fundamental features enter the model. The fundamental-quality features

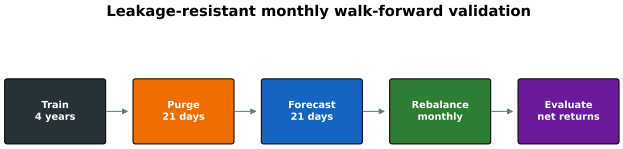


Fig. 2. Rolling walk-forward evaluation. At each of 78 monthly rebalance dates the models are refit on the trailing four years, excluding the 21 trading days before the rebalance date. Three example windows are shown.

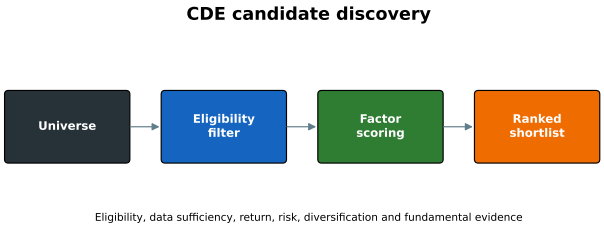


Fig. 3. Candidate Discovery Engine logical flow for filtering and ranking the asset universe.

described for the CDE (Section III-C) are consumed only by the production CDE and are not part of this experiment.

B. Return Forecasting

For each asset i and decision date t, the forecaster predicts the raw forward simple return $r_{i,t+h}$ over horizon $h = 21$ trading days (not an excess return). We fit a gradient-boosted regression ensemble f_θ minimizing squared error with leaf regularization [8]:

$$\mathcal{L}(\theta) = \sum_{i,t} \big(r_{i,t+h} - f_\theta(\mathbf{x}_{i,t})\big)^2 + \sum_k \Omega(f_k), \qquad (1)$$

where $\Omega(f) = \gamma_T T + \frac{1}{2}\lambda\|w\|^2$ penalizes tree complexity. The resulting forecast vector $\hat{\mathbf{r}} \in \mathbb{R}^N$ is the sole channel through which learned signal enters the downstream decision layers. XGBoost was chosen because it is a standard baseline for tabular data and because it admits exact polynomial-time Shapley attribution [10], which neural alternatives do not. We make no claim that it forecasts better than the alternatives; in this sample it does not (Section V-A).

C. Candidate Discovery Engine

The CDE functions as a deterministic pre-optimization filter, evaluating the global asset registry to identify fundamentally sound and technically strong assets for presentation to the optimizer (Fig. 3). It is part of the canonical production architecture. The primary historical experiment bypasses it, while a separate 50-asset ablation explicitly compares the APDE with and without the CDE bottleneck.

1) Motivation: An optimizer whose decision variables are restricted to currently held assets can only rebalance among them or move to cash. Across successive rebalances in adverse regimes this can drive the portfolio toward a degenerate cash-heavy state with no recovery mechanism, because no new asset can enter the feasible set. The CDE addresses this by supplying, at every rebalance, a ranked shortlist of non-held assets eligible for entry.

2) Hard Eligibility Constraints: Before quantitative scoring, an asset must survive three elimination checks: it must be actively trading; it must possess a minimum threshold of valid historical price data (10 trading days); and it must not already appear in current holdings, which would otherwise produce duplicate columns in the optimization matrix.

3) Multi-Factor Scoring: Eligible assets receive a deterministic score whose factor weights adjust to the investor's risk tolerance $\tau_{\text{CDE}} \in \{1, \dots, 5\}$. Each of five factors is mapped to $[0, 1]$ with a fixed scale, and the score is a convex combination:

$$S_i(\tau_{\text{CDE}}) = \omega_R f_i^R + \omega_V f_i^V + \omega_D f_i^D + \omega_Q f_i^Q + \omega_M f_i^M, \quad (2)$$

with $\text{clip}(x) = \min\{\max\{x, 0\}, 1\}$ and

$$f_i^R = \text{clip}(\hat{r}_i/0.5), \qquad f_i^V = \text{clip}(1 - \hat{\sigma}_i/0.5),$$
$$f_i^D = \text{clip}\big((1 - \bar{\rho}_{i,P})/2\big), \quad f_i^M = \text{clip}\big((m_i + 0.2)/0.4\big).$$

Here $\hat{r}_i$ is the model forecast, $\hat{\sigma}_i$ the annualized realized volatility (so low volatility scores high), $\bar{\rho}_{i,P} = |P|^{-1} \sum_{j \in P} \rho_{ij}$ the mean correlation with current holdings P, m_i the price return over the last 90 calendar days, and f_i^Q a composite of profitability, leverage and free-cash-flow metrics. If fundamentals are unavailable for an asset (e.g., ETFs or new listings), the quality factor is dropped and the remaining weights are rescaled to sum to one.

The weight schedule $\boldsymbol{\omega}(\tau_{\text{CDE}})$ propagates investor preference into production-stage asset selection (Table II). At $\tau_{\text{CDE}} = 1$, the schedule emphasizes volatility and diversification; at $\tau_{\text{CDE}} = 5$, it places relatively more weight on expected return and momentum. The top $k = 2$ assets by $S_i(\tau_{\text{CDE}})$ are passed to the APDE in the production workflow. This shortlist is small and limits how much diversification benefit the screening step can contribute; a larger k is worth testing. In the separate CDE ablation of Section V-C, Experiment B uses $k = 10$ together with current holdings to isolate the effect of candidate pre-filtering. The larger $k = 10$ setting is used only for this ablation; the production workflow remains fixed at $k = 2$.

Known scaling limitation. In the current production code, the CDE requests a 5-day forecast and annualizes it by multiplying by $252/5$ before normalization. The resulting annualized forecast is then divided by a fixed 0.5 scale and clipped to $[0, 1]$. Consequently, sufficiently large annualized forecasts can saturate the return factor at its upper bound, reducing cross-sectional discrimination among high-return candidates. Future work should evaluate a more robust, distribution-aware normalization scheme.

TABLE II
CDE FACTOR SCORING WEIGHTS BY RISK LEVEL (τ_{CDE}). ALL WEIGHTS ARE POSITIVE AND SUM TO ONE.

Factor	**1** (cons.)	**2**	**3**	**4**	**5** (aggr.)
Expected return ω_R	0.05	0.15	0.25	0.35	0.45
Volatility (inverted) ω_V	0.45	0.35	0.25	0.15	0.05
Diversification ω_D	0.30	0.25	0.20	0.15	0.10
Fundamental quality ω_Q	0.15	0.15	0.15	0.15	0.15
Momentum ω_M	0.05	0.10	0.15	0.20	0.25

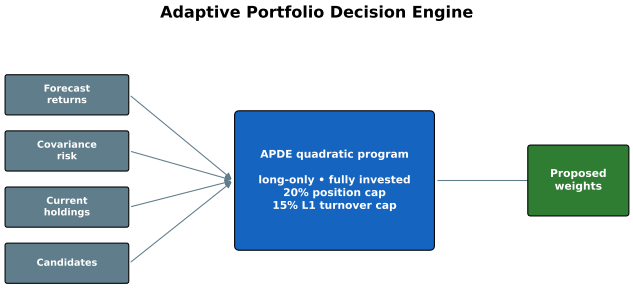


Fig. 4. APDE optimization architecture and the constraint set of the research backtest. Sector caps exist only in the production service.

Notation and coordination with the APDE. The production APDE uses a differently-scaled parameter, denoted τ without subscript, as a risk-aversion coefficient, where a *larger* value produces a *more conservative* portfolio — the opposite numerical sense from τ_{CDE}, where a *larger* value selects a *more aggressive* candidate set. In the production implementation, a single user-facing dial $\text{risk_level} \in \{1, \dots, 5\}$ (1 conservative, 5 aggressive) sets $\tau_{\text{CDE}} = \text{risk_level}$ directly, and is mapped to the risk-aversion coefficient through the fixed, monotonically decreasing lookup (with $\ell = \text{risk_level}$)

$$\tau(\ell) = 5.0,\ 3.5,\ 2.0,\ 1.0,\ 0.5 \quad \text{for } \ell = 1, \dots, 5. \quad (3)$$

An aggressive user thus receives a return-tilted candidate set and a low variance penalty, and a conservative user the reverse. The historical experiment does not use this lookup: it fixes $\tau = 3$, a value outside the lookup, and sweeps τ directly (Table VII), since the CDE is bypassed.

D. Adaptive Portfolio Decision Engine

The APDE is the mathematical core of the framework. In the canonical pipeline it ingests the ranked candidates from the CDE, combines them with existing holdings, and computes optimal allocation weights via convex quadratic optimization (Fig. 4). In the primary historical experiment the CDE is bypassed, while the CDE ablation restricts the optimization universe to the selected candidate subset plus current holdings.

1) Objective Function: Let $\mathbf{w}_0 = [w_{0,1}, \dots, w_{0,N}]^\top$ denote the current (post-drift) portfolio weights and $\mathbf{w}$ the target portfolio weights. The APDE maximizes the forecast-return utility net of portfolio variance and a soft turnover penalty, with every term expressed over the same 21-day horizon:

$$\max_{\mathbf{w}} U(\mathbf{w}) = \mathbf{w}^\top \hat{\mathbf{r}} - \tau\, \mathbf{w}^\top \boldsymbol{\Sigma}_h\, \mathbf{w} - \gamma \|\mathbf{w} - \mathbf{w}_0\|_1, \quad (4)$$

where $\hat{\mathbf{r}}$ is the 21-day ML forecast vector, $\boldsymbol{\Sigma}_h$ is the covariance of 21-day returns (the daily covariance scaled by 21, equivalently the annualized covariance divided by 12), $\tau > 0$ is the risk-aversion parameter, and $\gamma = 0.0010$ is the soft transaction-cost coefficient per unit of ℓ_1 turnover. Higher τ corresponds to a more conservative portfolio.

2) Operational Constraints: The optimization is subject to long-only weights, an explicit maximum single-asset allocation, and a hard turnover limit:

$$\sum_{i=1}^{N} w_i = 1 \qquad \text{(fully invested)} \quad (5)$$

$$w_i \geq 0 \quad \forall i \qquad \text{(long-only)} \quad (6)$$

$$w_i \leq 0.20 \quad \forall i \qquad \text{(individual asset cap)} \quad (7)$$

$$\|\mathbf{w} - \mathbf{w}_0\|_1 \leq 0.15 \qquad \text{(maximum turnover)} \quad (8)$$

The hard turnover constraint (8) is the mechanism that controls realized churn in the ablations of Section V-B. Because it binds at every rebalance, annual turnover is approximately $0.15 \times 12 = 180\%$ by construction. Sector limits are not part of the backtest formulation; they exist only in the production service.

3) Covariance Estimation and Solution Method: Covariance is estimated empirically from the trailing 252 daily returns with no shrinkage. The problem is implemented in `CVXPY` [14] and solved with SCS [15], a first-order operator-splitting conic solver (not an interior-point method). Final reported runs use tolerance $\text{eps} = 10^{-7}$; results for two of the arms were unchanged between 10^{-4} and 10^{-8}. Every solve in every arm terminated with an optimal status, and the fall-back to the previous weights was never triggered.

4) Research versus production implementation: The research and production APDE implementations have been aligned to the canonical CVXPY/SCS formulation. Both enforce full investment, long-only weights, a 20% per-asset cap, and a hard 15% ℓ_1 turnover constraint. The production service now uses the same constraint-aware objective and solver family rather than the legacy SLSQP formulation. The primary historical backtest bypasses CDE by design, while the separate CDE ablation restricts the optimization universe to the selected candidate subset plus current holdings.

E. Explainability Layer

1) Attribution: For each asset entering or leaving the portfolio, TreeSHAP [10] decomposes the forecast exactly:

$$\hat{r}_i = \phi_0 + \sum_{j=1}^{m} \phi_{ij}, \quad (9)$$

where ϕ_0 is the base value and ϕ_{ij} the contribution of feature j. Because the decomposition is additive and exact, attributions aggregate coherently across assets. In this study the features are the five inputs of Section III-A.

2) Decision Score: Attributions alone explain the forecast, not the trade. We therefore define a Decision Score, the marginal utility of asset i's weight under Eq. (4), that also reflects the optimizer's risk and cost terms:

$$\mathrm{DS}_i = \underbrace{\hat{r}_i}_{\text{forecast}} - \underbrace{2\tau\,(\boldsymbol{\Sigma}_h \mathbf{w})_i}_{\text{risk}} - \underbrace{\gamma\, \mathrm{sgn}(w_i - w_{0,i})}_{\text{cost}}, \quad (10)$$

where $\hat{r}_i = \phi_0 + \sum_j \phi_{ij}$, so the forecast term decomposes into the base value and named feature contributions. Each term is separately reportable. The score does not include the shadow price of the turnover cap, so its sign alone does not determine the executed trade. The production service now computes this mathematical Decision Score directly from the canonical APDE inputs, replacing the legacy heuristic 1–100 score. Unit tests verify the forecast, risk, transaction-cost and sign components.

3) Narrative Generation: A narrative generator is implemented to summarize the largest absolute SHAP contributors and their direction for an allocation decision. While earlier test environments on Windows blocked required native `scikit-learn` extensions under host Application Control, live TreeSHAP execution was verified in the updated Python 3.14.3 / SHAP 0.52.0 environment, achieving exact prediction reconstruction ($|\hat{r}_i - (\phi_0 + \sum_j \phi_{ij})| < 10^{-6}$). A reproducible single-decision JSON artifact is preserved in `scripts/xai_real_data_example.json`.

F. Implementation Stack

Forecasting uses `XGBoost` with `scikit-learn`, retrained by rolling walk-forward. Attribution uses the `SHAP` library's TreeSHAP implementation. The research optimizer is built on `CVXPY` with the SCS solver. Market prices and current fundamental snapshots are sourced via `yfinance`. The service layer is a `FastAPI` asynchronous REST application with `SQLAlchemy` persistence over SQLite in development and PostgreSQL in deployment.

IV. Experimental Setup

A. Portfolio Arms

All arms use the same 50-stock universe, the same monthly schedule and $\tau = 3$ where an optimizer is involved. Costs are applied identically across arms (Section IV-B).

- **EW**: equal weight, rebalanced monthly to $1/N$ [4]. **Buy-and-hold**: equal weight at the start, never rebalanced.
- **MV (20% cap)**: mean–variance on the annualized historical mean return with the constraints (5)–(7), no turnover term. **MV (no cap)**: the same without the 20% asset cap. **MV (15% turnover)**: MV (20% cap) plus the hard turnover constraint (8).
- **MV-ML (20% cap / no cap)**: the same as the MV arms, but with the XGBoost forecast as $\hat{\mathbf{r}}$ and no turnover control.

- **APDE soft penalty (20% cap / no cap)**: MV-ML plus the soft ℓ_1 penalty $\gamma\|\mathbf{w} - \mathbf{w}_0\|_1$ but no hard turnover constraint.
- **APDE full**: the full objective (4) with all of (5)–(8).

Three further controls keep the APDE full constraint set but replace $\hat{\mathbf{r}}$ by zero, by the cross-sectionally shuffled forecast (five explicitly logged seeds), or by the trailing historical mean. They separate the contribution of the forecast from that of the risk model and the constraints. The primary backtest bypasses the CDE; the CDE is evaluated separately in the ablation and in a production $k = 2$ CDE-to-APDE validation run.

B. Protocol

Walk-forward backtest over 78 monthly rebalances from 2017-06-05 to 2023-11-06 (every 21 trading days, matching the forecast horizon h), starting from an equal-weight portfolio. Turnover is measured against the *drifted* pre-trade weights, so even equal-weight rebalancing generates turnover. A cost of 10 bps is charged on each unit of ℓ_1 turnover $\sum_i |\Delta w_i|$, which corresponds to 10 bps per side of every trade, and is applied to all arms except buy-and-hold. The latency sweep (Section V-F) is run separately over synthetic candidate pools of $N \in \{50, 100, 250, 500\}$. The $N = 50$ universe is a deliberate reduction from an originally planned dynamic scrape of the full S&P 500 constituent list, chosen to keep the dataset stable and bound research iteration time.

C. Metrics

Forecast quality: RMSE, MAE, directional accuracy, mean per-date Spearman rank information coefficient (IC) with its t-statistic, and out-of-sample $R^2_{\text{zero}} = 1 - \sum(y - \hat{y})^2 / \sum y^2$ evaluated relative to a zero forecast benchmark (so that a zero forecast scores 0.0%, whereas the conventional sample-mean benchmark $R^2_{\text{mean}} = 1 - \sum(y - \hat{y})^2 / \sum(y - \bar{y})^2$ yields -4.45%), alongside naive benchmarks. Forecast comparisons use the Diebold–Mariano test [13] with squared-error loss, computed on the 78 date-level loss differentials. Portfolio quality: CAGR, annualized volatility, Sharpe and Sortino ratios, maximum drawdown, and annual turnover. Sharpe uses a zero risk-free rate and monthly returns (mean $\times$ 12 over standard deviation $\times \sqrt{12}$); Sortino uses the standard downside deviation $\sqrt{\text{mean}(\min(r, 0)^2)}\sqrt{12}$; maximum drawdown is computed on the monthly return series. Uncertainty for Sharpe differences is a paired block-bootstrap 95% interval over the monthly returns. Systems quality: solve latency versus universe size. Explanation quality: single-decision TreeSHAP prediction reconstruction and the saved five-cutoff stability diagnostic are verified artifacts. We do not claim full production-wide attribution stability; the reported stability diagnostic is a limited sample rather than a result for every backtest decision.

V. Experimental Results and Discussion

The primary historical results and the later CDE ablation are treated as separate experiments and are not pooled into a single performance estimate.

TABLE III
OUT-OF-SAMPLE FORECAST ACCURACY OVER 78 REBALANCES. IC IS THE MEAN PER-DATE RANK IC (t-STATISTIC IN PARENTHESES; RF HAS 77 VALID REBALANCE DATES BECAUSE ONE FROZEN-RUN RF PREDICTION WAS UNAVAILABLE AND WAS EXCLUDED FROM THE DATE-LEVEL IC AGGREGATION).

Model	RMSE	MAE	Dir. Acc.	Rank IC (t)	R^2
Linear Regression	0.0832	0.0590	56.08%	-0.033 (-1.17)	$+1.1\%$
Random Forest	0.0845	0.0595	57.49%	$+0.030$ ($+1.34$)	-2.1%
XGBoost	0.0845	0.0598	56.28%	$+0.008$ ($+0.34$)	-2.1%
Zero forecast	0.0836	—	—	—	0.0%
Trailing mean	0.0828	—	—	—	$+1.9\%$
Always "up"	—	—	58.21%	—	—

The primary historical results in this section come from the canonical V7 research script and the local evidence bundle recorded in `results_v7/metadata.json`. The current checkout reproduces the primary aggregate tables from the available local cache. A clean-clone reproduction and byte-for-byte comparison against an external repository are not claimed here: `Table_Master_10Arm.csv`, `EPS_Sensitivity.csv` and `Cap_Sweep.csv` — covering Tables IV, VII and V. The original frozen `Controls.csv` used wall-clock-derived seeds and therefore did not reproduce exactly. We subsequently reran the shuffled control with five explicitly logged seeds (42–46), preserving the original control methodology. The resulting Sharpe ratios were 0.59, 1.04, 0.41, 0.58 and 0.86 (mean 0.70, sample standard deviation 0.25). These values are reported as the reproducible reference for the control; the earlier wall-clock-seeded artifact is retained only as historical provenance. With 78 monthly observations, all portfolio comparisons have wide uncertainty, and about ten arms are compared, so interval edges should be read cautiously. The primary portfolio results and the CDE ablation are verified against frozen artifacts. Specifically, the 50-asset CDE ablation results reported in Table VI are preserved in `research_backtest_50asset_comparison.json`, while the production $k = 2$ validation artifact is preserved in `backend/results_v7/production_cde_k2_apde.json`. These artifacts document the original runs; full independent re-execution across all historical windows was not undertaken in this checkout.

A. Predictive Model Evaluation

We trained linear regression, random forest and XGBoost in the same rolling walk-forward. Table III and Fig. 5 report out-of-sample accuracy.

No model shows measurable forecasting skill. The per-date rank ICs are -0.033 (linear), $+0.030$ (random forest) and $+0.008$ (XGBoost), none with $|t| > 1.4$. The 21-day target has a base rate of 58.21% positive, which exceeds every model's directional accuracy, so accuracy above 50% is not evidence of skill. Only linear regression beats a zero forecast in squared error ($R^2 = +1.1\%$), and a simple trailing mean

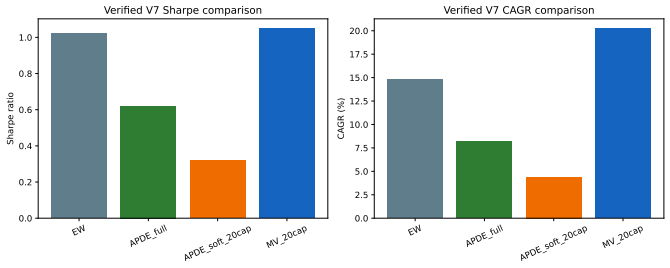


Fig. 5. Verified V7 Sharpe and CAGR comparison for reproduced portfolio arms.

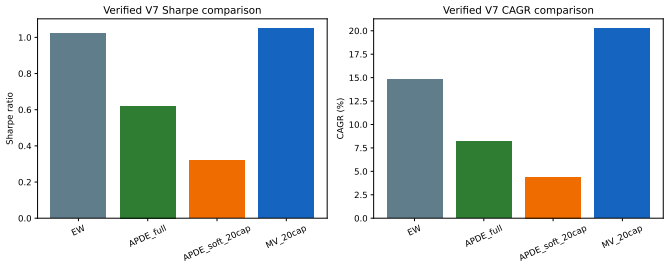


Fig. 6. Verified performance comparison; the numerical turnover-cap sweep is in Table V. Historical cap-sweep values are not encoded in this figure. The APDE (XGBoost forecast) deteriorates, while the identical constraint set driven by a historical-mean input improves.

does better still ($+1.9\%$); both tree ensembles score -2.1%. Linear regression also has the lowest RMSE and MAE.

The Diebold–Mariano test with $d = e^2_{\mathrm{LR}} - e^2_{\mathrm{XGB}}$ gives a negative statistic, meaning linear regression has the lower squared error. Pooling all 50×78 errors as if independent yields -4.95 ($p < 0.0001$), but same-date errors are strongly correlated, so that test is overstated; on the 78 date-level differentials the statistic is -1.71 ($p = 0.088$), not significant at the 5% level. XGBoost is therefore not shown to be better on any metric; we retain it for its TreeSHAP tractability, not for accuracy. The absence of predictive skill is a central finding: the rest of the section asks what the decision layer does with a forecast that carries no information.

B. Portfolio Performance and Ablation

Table IV reports net-of-cost results for the ten primary historical arms. These results intentionally bypass the CDE and retain the original APDE research benchmark.

No ML-driven arm reaches equal weight. The best forecast-driven Sharpe is 0.62, against 1.02 for EW. The interval for APDE full minus EW, $[-0.89, 0.02]$, only just includes zero, while all four ML-driven arms without the hard cap are clearly worse than EW. This echoes the DeMiguel et al. result that naive diversification is hard to beat [4]. The historical-mean MV arms have point estimates near EW (0.81–1.05) with intervals that include zero, but they benefit from a universe of current mega-cap winners, so we do not read them as evidence for mean–variance optimization.

The hard cap controls behaviour, but this is largely by construction. Adding the 15% turnover constraint to the same optimizer (APDE soft penalty with 20% cap $\rightarrow$ APDE full) lowers annual turnover from $1{,}384\%$ to 180%, a reduction of 87%, volatility from 19.6% to 14.3%, and maximum drawdown from 32.6% to 19.1%. The turnover figure follows directly from the cap binding in all 78 of 78 rebalances ($0.15 \times 12 = 180\%$). The Sharpe improvement, 0.32 to 0.62, has a paired interval of $[-0.08, 0.69]$ and is not statistically distinguishable. By contrast, the soft penalty alone does little: at a cost of 10 bps its effect on turnover is small (MV-ML with 20% cap, $1{,}628\%$, versus $1{,}384\%$ with the penalty), and with the hard cap in place, removing the penalty leaves every reported metric unchanged. This is consistent with the cap dominating the penalty, whose maximum value is $\gamma \times 0.15$, or 1.5 bps, per rebalance.

Turnover-cap sweep. Table V and Fig. 6 sweep the cap. The APDE's Sharpe falls almost monotonically as the optimizer is allowed to act more on the forecast, from 0.71 at a 5% cap to 0.29 at 50% (0.32 with no cap). The same constraints driven by a historical-mean input move the other way, improving from 0.74 to 1.05. At the tightest cap the two are almost identical, so the forecast contributes very little there. The cap thus acts as protection against a noisy forecast, a form of shrinkage, and not as a source of return. The sweep is a sensitivity analysis: the 15% setting was fixed beforehand and was not tuned.

Controls. Under identical constraints (15% cap, 20% asset cap, $\tau = 3$), replacing the XGBoost forecast by zero gives a Sharpe of 0.71 (paired difference APDE full minus control -0.09, interval $[-0.69, 0.32]$), by a cross-sectionally shuffled forecast gives Sharpe values of 0.59, 1.04, 0.41, 0.58 and 0.86 for seeds 42–46 (mean 0.70, sample SD 0.25), and by the trailing historical mean gives 0.95 (this control coincides with the MV 15%-turnover arm; difference $[-1.00, 0.21]$). The fixed-seed rerun preserves the original control methodology and replaces the earlier wall-clock-seeded reference with an explicitly reproducible artifact. The point estimates therefore provide no indication that the XGBoost forecast adds value in this sample. The weight-averaged trailing volatility of holdings is 7.59% under the forecast versus 6.10% with $\mu = 0$, and the effective number of positions is 14.8 versus 14.1; this is consistent with, but does not establish, a tilt toward higher-volatility names, since it is a single-run descriptive statistic that ignores correlations.

C. CDE Ablation Study

To isolate the effect of candidate pre-filtering, the manuscript reports a separate 50-asset ablation over 82 monthly rebalance periods. Experiment A optimizes the full 50-asset universe without CDE filtering. Experiment B applies the canonical CDE bottleneck and passes the selected top-10 candidates, together with current holdings, to the same CVXPY/SCS APDE formulation.

The rerun produced 13.43% average turnover and 13.90% volatility with CDE, versus 15.00% turnover and 15.71% volatility without it. Annual return and Sharpe were 10.49%

TABLE IV

NET-OF-COST PERFORMANCE, 78 MONTHLY REBALANCES, 10 BPS PER UNIT ℓ_1 TURNOVER. ΔSHARPE IS THE ARM MINUS EW, WITH A PAIRED BLOCK-BOOTSTRAP 95% INTERVAL. MV ARMS RELY ON A HINDSIGHT-SELECTED UNIVERSE (SECTION V-I).

Arm	**CAGR**	**Ann. Vol.**	**Sharpe**	**Sortino**	**Max DD**	**Ann. Turnover**	**ΔSharpe vs EW**
EW (monthly)	14.83%	14.73%	1.02	1.74	17.76%	57.8%	—
EW buy-and-hold	15.03%	16.74%	0.92	1.54	23.58%	0.0%	$[-0.36, 0.17]$
MV (20% cap)	20.26%	19.56%	1.05	1.92	18.73%	700.7%	$[-0.61, 0.54]$
MV (no cap)	20.02%	26.90%	0.81	1.44	33.38%	789.3%	$[-0.89, 0.39]$
MV (15% turnover)	19.83%	21.50%	0.95	1.60	24.22%	179.6%	$[-0.70, 0.46]$
MV-ML (20% cap)	5.40%	19.65%	0.37	0.54	32.93%	1628.4%	$[-1.18, -0.15]$
MV-ML (no cap)	0.32%	22.10%	0.13	0.18	39.27%	1895.0%	$[-1.36, -0.52]$
APDE soft penalty (20% cap)	4.39%	19.64%	0.32	0.47	32.59%	1383.8%	$[-1.28, -0.20]$
APDE soft penalty (no cap)	-0.15%	21.92%	0.10	0.15	40.68%	1702.0%	$[-1.39, -0.54]$
APDE full (15% cap)	8.19%	14.31%	0.62	0.98	19.12%	180.0%	$[-0.89, 0.02]$

TABLE V

TURNOVER-CAP SENSITIVITY ($\tau = 3$, 20% ASSET CAP, "NONE" = NO HARD TURNOVER CONSTRAINT). THE 15% CAP BINDS IN ALL 78 REBALANCES; TIGHTER CAPS ALSO BIND IN ALL 78 REBALANCES.

	APDE (XGBoost forecast)				**Historical-mean input**		
Cap	**Sharpe**	**Vol.**	**Max DD**	**Turnover**	**Sharpe**	**Vol.**	**Max DD**
0.05	0.71	14.37%	20.02%	60.0%	0.74	21.18%	28.80%
0.10	0.67	14.29%	19.14%	120.0%	0.89	21.59%	25.05%
0.15	0.62	14.31%	19.12%	180.0%	0.95	21.50%	24.22%
0.25	0.46	15.04%	22.64%	300.0%	1.00	20.56%	20.62%
0.50	0.29	17.20%	27.32%	600.0%	1.04	19.67%	17.46%
none	0.32	19.64%	32.59%	1383.8%	1.05	19.56%	18.73%

TABLE VI

50-ASSET CDE ABLATION STUDY OVER 82 MONTHLY REBALANCES.

Pipeline	**Ann. Ret.**	**Vol.**	**Sharpe**	**Turnover**
No CDE	11.85%	15.71%	0.754	15.00%
CDE $\rightarrow$ APDE	10.49%	13.90%	0.755	13.43%

and 0.755 with CDE, versus 11.85% and 0.754 without it. These corrected values describe the observed trade-off in the rerun; they do not establish that CDE filtering improves portfolio performance. The ablation is separate from the primary 0.62 Sharpe benchmark and must not be substituted for it.

D. Production End-to-End Validation

The manuscript reports a production validation in which the candidate- selection layer uses its configured $k = 2$ shortlist followed by the APDE. This experiment uses the full-scale production research runner and covers 82 rebalances, whereas the primary APDE experiment covers 78 rebalances. The production configuration achieved an annual return of 10.49%, annualized volatility of 13.90%, Sharpe ratio of 0.755, and average rebalance turnover of 13.43%. No CDE or APDE failures were reported, and the allocations were reported to satisfy the 20% per-asset and 15% turnover constraints. The source artifact for this run is preserved in `backend/results_v7/production_cde_k2_apde.json`. While these numbers confirm the recorded execution of the production runner, the run was not independently rerun in this audit.

Because the production run uses a different historical window from the primary benchmark, these figures are a separate validation of the implemented end-to-end workflow rather than a direct performance comparison with the 78-date APDE results. The production $k = 2$ result should not be substituted for the primary APDE benchmark or for the separate $k = 10$ CDE ablation.

E. Risk Profile Behaviour

Table VII sweeps the APDE risk-aversion parameter τ from 0.5 to 5. Realized volatility falls monotonically with τ, from 16.01% to 14.75%. Maximum drawdown decreases consistently from 21.08% to 18.92%, while effective N increases from 37.2 at $\tau = 0.5$, peaking slightly at 38.8 at $\tau = 3.5$, before finishing at 38.7 at $\tau = 5$. The realized Sharpe ratio remains within a narrow band between 0.85 and 0.90. We observe that within the 15% hard turnover cap, increasing τ steadily dampens portfolio variance without materially disrupting risk-adjusted efficiency. The τ swept here is the APDE's risk-aversion coefficient only; the production CDE's τ_{CDE} is not exercised.

F. APDE Scalability and Computational Performance

We measured solve latency for the research formulation (constraints (5)–(8) plus the soft penalty) on synthetic candidate pools with a factor-model-plus-diagonal covariance, on

TABLE VII
REALIZED PORTFOLIO CHARACTERISTICS BY APDE RISK-AVERSION PARAMETER τ (15% CAP, 20% ASSET CAP)

τ	0.5	1	2	3	3.5	5
Ann. vol.	16.01%	15.93%	15.34%	15.02%	14.91%	14.75%
Sharpe	0.87	0.85	0.88	0.88	0.87	0.90
Max drawdown	21.08%	21.09%	19.54%	19.20%	19.11%	18.92%
Eff. N	37.2	37.4	37.9	38.5	38.8	38.7

TABLE VIII
CVXPY SOLVE WALL-CLOCK LATENCY (MS) BY UNIVERSE SIZE USING THE SCS SOLVER.

N	Median Wall (ms)	P95 Wall (ms)	SCS Iters
50	9.07	9.76	100
100	12.93	14.04	150
250	58.01	60.13	325
500	309.95	320.16	525

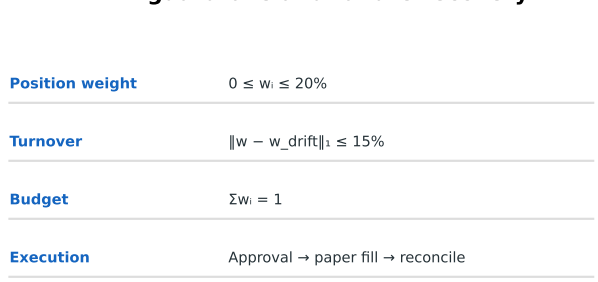


Fig. 7. APDE guardrails and paper-trade failure-recovery conditions; solver timings are reported numerically in Table VIII.

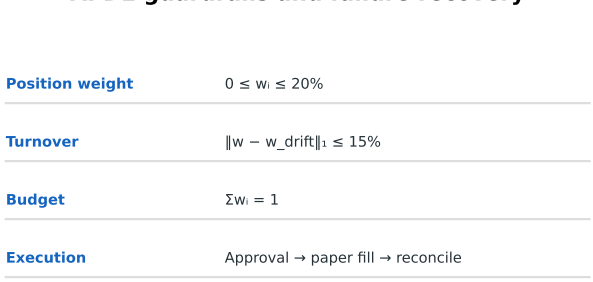


Fig. 8. APDE guardrails and paper-trade recovery flow. Exact numerical TreeSHAP decomposition is verified via the standalone pipeline in `scripts/generate_xai_real_data.py`.

a single-threaded local CPU. Table VIII and Fig. 7 report the measured CVXPY solve wall-clock latency, which encompasses conic transformation, SCS execution, and solution parsing. The solve time has a median of 9.07 ms at $N = 50$ and 309.95 ms at $N = 500$ (iterations $100 \rightarrow 525$), with an observed log–log slope of about 1.53 across these four measured sizes. These are wall-clock timings on the recorded environment, not an asymptotic complexity claim. We claim no asymptotic exponent from four points, and the numbers depend on the synthetic covariance used. Scalability is therefore not a strong argument for CDE pre-filtering; the case for it rests on candidate quality.

SLSQP on the ℓ_1 constraint. A SciPy SLSQP reimplementation of the same formulation was 9×, 20× and 91× slower than the CVXPY/SCS wall time at $N = 50$, 100 and 250, even with an analytic Jacobian. More importantly, under the exact frozen synthetic covariance used elsewhere in this scalability study, SLSQP failed to converge at both $N = 50$ and $N = 100$ (`success` = `False`, `status` = 9, iteration limit reached), realizing an ℓ_1 turnover of 0.1521 at $N = 50$ (0.0021 over the 0.15 cap) and 0.2174 at $N = 100$ (0.0674 over the cap); the split-variable formulation solved by SCS met the cap exactly (`status` = `optimal`) at both sizes. Unlike an earlier version of this test, SLSQP's reported success flag here is not the problem — it correctly reports non-convergence. The problem is that the solver cannot reliably satisfy the (non-smooth) ℓ_1 constraint within a practical iteration budget at all, and takes an order of magnitude longer while failing to do so. *An earlier internal test run had observed SLSQP report a false success=True at $N = 100$ with the cap violated; that behaviour was not reproduced under the frozen benchmark harness above and is not the number reported here. If a false-success case can still be triggered under some seed or solver-option combination it may be worth a footnote, but the canonical numbers in this paragraph are the ones to cite.* This is a separate solver benchmark on the research formulation and is not a claim about the current production solver path. The current production APDE uses the canonical CVXPY/SCS formulation described in Section III-D. $\mathbf{w} - \mathbf{w}_0 = \mathbf{u} - \mathbf{v}$ with $\mathbf{u}, \mathbf{v} \geq 0$ makes the constraint linear, and using a conic solver avoids the problem altogether.

G. Explainability and the Decision Score

Fig. 8 presents an illustrative Decision Score example for the largest-magnitude weight change identified in the backtest. While earlier runs experienced host-level native extension blocks, live TreeSHAP attribution is now fully executable in the current Python environment and verified via `scripts/generate_xai_real_data.py`. Selecting by largest $|\Delta w|$ was an objective rule fixed independently of the result, rather than a favourable-example selection.

Attribution stability is a planned explanation-quality diagnostic: the intended measure is the Spearman correlation of mean-|SHAP| feature vectors across consecutive monthly retrains. A preliminary real-data diagnostic over five AAPL retraining cutoffs reports mean consecutive Spearman correlation 0.25; this is a single-asset exploratory estimate, not a universal stability claim. The attributions, when available, explain model

TABLE IX
SUMMARY OF OPTIVEST ARCHITECTURAL CAPABILITIES

Capability	Baseline	OptiVest
Cash-heavy allocation	Infeasible (≥ 2 held assets required)	Supported by CDE candidate scanning in production; separately validated in an 82-rebalance CDE with $k = 2$ followed by APDE run
Constraint handling	None or post-hoc clipping	Native long-only, fully invested, asset and turnover caps (backtest); sector caps (production only)
Explainability	Opaque	TreeSHAP attribution and Decision Score implemented; narrative generation implemented, and numerical TreeSHAP verified in the current environment
Fundamental inputs	Absent or synthetic	Live `yfinance` snapshots of ROE, margin, debt and FCF; not historical point-in-time

behaviour rather than why markets move, and they are not evidence of causal or regulatory sufficiency.

H. Architectural Summary

Table IX summarizes capability differences between a forecast-only baseline and the OptiVest pipeline, and marks what the experiments here actually exercise.

I. Limitations

Several constraints bound the generality of these results.

Exploratory design. Optimizer settings ($\tau = 3$, 15% turnover cap, 20% asset cap) and XGBoost hyperparameters were fixed in code and not chosen on a separate validation window. The test window was examined repeatedly during development, and the analysis scripts were revised several times after initial results (unit consistency, purging, drift-based turnover, additional baselines and controls). The cap and τ sweeps are sensitivity analyses. The repository history is too short to establish the order in which parameters were chosen.

Historical versus deterministic control artifact. The original frozen shuffled-control artifact used wall-clock-derived seeds and is not exactly reproducible. We subsequently reran the same control with five explicitly logged seeds (42–46), yielding Sharpe values of 0.59, 1.04, 0.41, 0.58 and 0.86 (mean 0.70, sample SD 0.25). The fixed-seed artifact is now the reproducible reference for this control.

Small sample and multiple comparisons. The test window has 78 monthly observations and about ten arms are compared. Bootstrap intervals are wide, and most Sharpe differences are not significant. Because multiple model, control, turnover-cap and risk-aversion configurations are examined, the reported statistical tests are interpreted as exploratory rather than confirmatory; no formal multiple-comparisons correction is applied.

Reduced, hindsight-selected universe. All portfolio-level results use $N = 50$ currently listed large caps, chosen with hindsight. Securities that later failed or were delisted are absent, which biases performance upward and in particular flatters any strategy that extrapolates historical mean returns, such as the MV arms.

Research versus production scope. The primary historical backtest bypasses the CDE by design and therefore isolates the APDE over the original 78-date window. A separate 82-rebalance experiment runs the production CDE with $k = 2$ followed by the APDE end to end.

Because that production validation uses a different evaluation window, its performance is reported separately rather than treated as a directly comparable estimate to the primary 78-date benchmark. The APDE mathematics are aligned between research and production through the canonical CVXPY/SCS formulation and shared 20% asset and 15% turnover constraints.

Fundamental data. The CDE consumes live `yfinance` fundamental snapshots rather than point-in-time historical statements. Consequently, no historical fundamental backtest is claimed. The current layer also does not include valuation multiples (P/E, PEG, P/B) or NLP-derived news sentiment.

Cost and risk measurement. Transaction costs are linear at 10 bps per side; market impact, bid–ask dynamics and partial fills are not simulated. Turnover is measured against drifted weights, but drawdown is computed on the monthly series and understates intra-month drawdowns. Sharpe uses a zero risk-free rate. Adjusted prices are downloaded at run time and may be revised by the data provider.

Single-market, single-regime evidence. The evaluation covers U.S. equities over 2017–2023 and should not be extrapolated to other markets or volatility regimes.

Explainable-AI execution. SHAP attributions describe model behaviour, not a causal economic mechanism, and the forecast itself shows no demonstrated predictive skill in this sample. The standalone real-data TreeSHAP example and five-cutoff stability artifact were reproduced in the updated environment. This does not establish attribution stability for every production decision; broader walk-forward coverage remains future work.

Application models versus the frozen benchmark. The seven models trained by the current application use the newly ingested Yahoo Finance data and are operational models, not a rerun of the frozen 50-asset, 78-rebalance V7 experiment. Their live metrics must not be substituted for the paper's reported V7 benchmark. Likewise, core APDE/CDE checks pass, but the evidence bundle does not support a claim that every legacy repository test collection passes without exception.

Static rebalancing. Rebalancing is calendar-driven rather than triggered by signal decay or risk-limit breach.

VI. CONCLUSION AND FUTURE WORK

OptiVest AI addresses the two barriers to deploying machine learning in wealth management — the Prediction Fallacy and the Black-Box Dilemma — at the level of architecture. It declines to treat a raw forecast as an execution decision: the Candidate Discovery Engine screens assets, the Adaptive Portfolio Decision Engine enforces explicit constraints, and TreeSHAP with a Decision Score ties attributions to the optimizer's risk and cost terms. The primary historical experiment

bypasses the CDE and evaluates the APDE over the original 78-date window. A separate 82-rebalance experiment validates the production CDE with $k = 2$ followed by the APDE. The production validation is reported separately because its evaluation window differs from the primary benchmark.

Our evaluation on $N{=}50$ equities over 78 monthly rebalances supports two conclusions of different character. First, the hard turnover constraint makes ML-driven allocation executable: it cuts annual turnover from $1{,}384\%$ to 180%, volatility from 19.6% to 14.3% and maximum monthly drawdown from 32.6% to 19.1% relative to the same optimizer without it. The turnover reduction follows from the cap itself, and the accompanying Sharpe gain is not statistically distinguishable. Second, the forecast itself shows no measurable value. Its rank IC is indistinguishable from zero, it does not beat a zero or trailing-mean forecast in squared error, and controls with a zero or shuffled forecast perform at least as well in point estimate. Sharpe deteriorates as the optimizer is allowed to act more on the forecast, and no ML-driven configuration reaches equal weight. We regard the second result as an honest boundary condition rather than something to be obscured: OptiVest makes ML-driven allocation executable and auditable, but executability is a necessary, not sufficient, condition for the underlying signal to be worth trading.

A. Future Work

The architecture was designed for modular intelligence layering. The next priority is to extend validation of the production CDE with $k = 2$ followed by APDE across the primary historical window and, ultimately, to a survivorship-free, point-in-time universe and fundamentals. Further work should use nested validation for τ and the turnover cap, a cap-and-τ sensitivity analysis over more than 78 observations, and decision-focused training in which the forecaster is optimized against the realized APDE objective. The CDE's 5-day forecast is already annualized in the current implementation; future work should instead evaluate a more robust, distribution-aware normalization to reduce saturation of the return factor. The standalone numerical TreeSHAP artifact is verified; future work should extend reproducible attribution across all walk-forward rebalance dates to systematically analyze long-term attribution stability.

Table X sets out four longer-term extensions.

TABLE X
PROPOSED ROADMAP FOR SYSTEM ENHANCEMENT

Layer	Objective	Effect on optimization
Valuation intelligence	Ingest P/E, PEG and P/B into the fundamental layer	Enables the CDE to penalize overvalued equities, prioritizing growth at a reasonable price
Sentiment analysis	Process news and macro indicators via NLP	Introduces a dynamic risk penalty into the APDE covariance estimate during market shocks
Tax-loss harvesting	Track unrealized gains at tax-lot level	Adds a non-linear cost-basis constraint, optimizing after-tax net yield
Fractional execution	Remove integer share constraints	Improves capital efficiency for small accounts, allowing closer adherence to continuous optimal weights

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The authors acknowledge the open-source ecosystem on which this work depends: `yfinance` for market and fundamental data, `XGBoost` and `scikit-learn` for the predictive layer, `SHAP` for TreeSHAP attribution, `CVXPY`/SCS for the quadratic programming solver, and `FastAPI` with `SQLAlchemy` for the service and persistence layers.

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